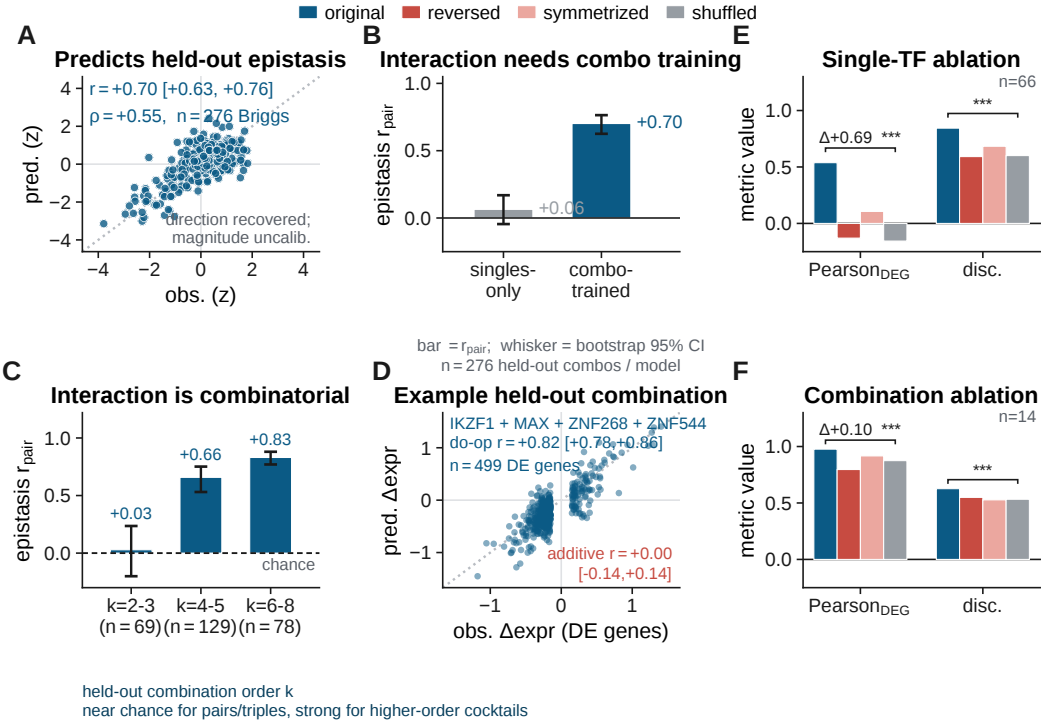
original
reversed
symmetrized
shuffled
A
Predicts held-out epistasis
r = +0.70 [+0.63, +0.76]
ρ = +0.55, n = 276 Briggs
direction recovered;
magnitude uncalib.
pred. (z)
obs. (z)
B
Interaction needs combo training
epistasis r_pair
+0.70
+0.06
singles-
only
combo-
trained
E
Single-TF ablation
n=66
metric value
Δ+0.69 ***
***
Pearson_DEG
disc.
bar = r_pair; whisker = bootstrap 95% CI
n = 276 held-out combos / model
C
Interaction is combinatorial
epistasis r_pair
+0.03
+0.66
+0.83
chance
k=2-3
(n = 69)
k=4-5
(n = 129)
k=6-8
(n = 78)
D
Example held-out combination
IKZF1 + MAX + ZNF268 + ZNF544
do-op r = +0.82 [+0.78,+0.86]
n = 499 DE genes
additive r = +0.00
[-0.14,+0.14]
pred. Δexpr
obs. Δexpr (DE genes)
F
Combination ablation
n=14
metric value
Δ+0.10 ***
***
Pearson_DEG
disc.
held-out combination order k
near chance for pairs/triples, strong for higher-order cocktails